DSA 104 Mock Exam:

ML and AI in Chemistry

Name:

Matr. No.:

Signature:

Please read the questions carefully. Draw and write down ONLY what is necessary to answer the question in the space allocated underneath each question. You use some extra paper for your preliminary drawings or sketches (can be requested), only the stapled document will be corrected. Please do not use pencil, Tipp-Ex and/or red pens.

Task	**Score**
Definitions (Q1-2)	/ 12
Data (Q3-7)	/ 17
Basic ML (Q8-14)	/ 37
NNs (Q15-20)	/ 21
Critical thinking (Q21-23)	/ 14
Case study (Q24)	/ 14
Total	**/ 115**

1) What is *machine learning*? Give a **definition** and **one example application**. Name at least **three advantages of ML**. (5P)

2) **Give a definition** for each of the following terms. (7P)

Feature matrix:

Label:

Classification:

Empiric Risk Minimisation:

Deep Learning:

Halluzination:

3) Data: What are typically some key limitations with data in the life sciences? **Name two examples** and **state how each affects any ML** models. (4P)

4) What is feature engineering? Give a brief **definition** and name at least **three fundamental tools**! (4P)

5) What is scaling needed for? Why is it best implemented in a pipeline? (2P)

6) **Explain** data leakage (in the technical sense, not in the media context), give **two examples**, and **name one fundamental practice** how to avoid data leakage. (4P)

7) **What is data bias**? State **two example**s where could they be introduced? (3P)

8) **Explain** underfitting, provide an **exemplary graph** and discuss **how to resolve it**! (3P)

9) Consider the following datatypes and **assess how they are usually represented**, and **name a challenge that might be associated with each** of them. **Name an example application (regression/classification) for each** and describe **which ML model** would be used for them. (12P)

- Protein sequence

- Molecular structure

- Process parameters (Temperature, pressure, etc.)

10) Why is accuracy alone not the best metric? **Explain**! Name at least **two more metrics** that could be used to judge a classification model. (3P)

11) **Explain** how a decision tree works. What is **a very common problem** of this model and **how can it be addressed**? What **advantages and disadvantages does a Random Forest** provide in comparison? **Name two each**! (7P)

12) **Explain Crossvalidation** and **why it is needed**. Name **two common methods** and **explain them briefly**. (6P)

13) How to train your ~~dragon~~ model: Give some pseudocode for a training pipeline of any ML model. (3P)

14) **Explain Exploration** and **Exploitation**. **Give an example** of where these principles may find application. (3P)

15) **Explain** how an artificial neuron **calculates its output from its inputs**. Explain **activation**. (3P)

16) What is an MLP / FNN? Give a **brief definition** and explain at least **three of its key components**. (4P)

17) **Explain the components** of a convolutional layer. Give **an example** where a CNN might be applied in the life sciences. (3P)

18) State **one reason** why transformer models are dominant for sequence learning? Name at least **one other model class** that may be applicable there and why these models are at a **disadvantage** compared to transformers. (3P)

19) What is transfer learning? Give a **definition**! **Name and explain** in short **two approaches**. (5P)

20) Explain in short how a GNN works (3P)

21) Name **three key ethical aspects** of machine learning (3P)

22) **Explain** applicability domains and **name and explain at least two different AD metrics** and explain in short. (5P)

23) Critical thinking – comment on the following statements: (6P)

- “More complex models and a high amount of data lead to better insights/predictions”
- “ChatGPT suggested a new catalyst (SMILES) and it actually worked. It is great to see that the finally implemented some chemistry knowledge in LLMs!”
- “With the progress in AI and automation, soon, there will no longer be a need for human scientists.”

24) Case study: You are supposed to predict the activity of some small molecules against a certain cancer cell line.

a) Propose a high-level ML pipeline (data → model → validation → outcome). **Explain each part** in short and describe what **considerations for each step** may be valid (10P)

b) Identify two major challenges specific to this task. (2P)

c) Explain how you would evaluate whether the model is clinically useful. (2P)